Sahaj Bhadja

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Summary

Software Engineer with 2+ years building web, data, and AI-powered products in **Java**, **Python**, **JavaScript**, and C++, owning the full lifecycle from design and code review to testing, debugging, deployment, and documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Improved **performance** of a multi-tenant education platform for 500+ users by redesigning schemas and indexed query patterns, cutting dashboard latency from 1.2s to 350ms in production for every tenant.
- Restructured fault-tolerant asynchronous AI job processing with explicit state transitions and failure isolation, tripling **throughput** across 1K+ jobs and shipping through CI/CD at 99%+ reliability across 5K+ monthly jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a customer-facing React and Python Flask application on AWS with **multimodal AI** models for upload, rubric, and grading workflows, serving 150+ instructors and 10K+ daily requests.
- **Debugged** OCR timeouts and model failures in production, adding fallback routing and multithreaded batch processing that kept the product reliable at peak load and cut feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** with data pipelines over 100K+ autonomous-driving trajectory logs, improving long-horizon trajectory planning 18% across closed-loop simulation runs of driving scenarios.
- Designed a multi-head attention encoder and **validated** it against three baselines with documented failure modes, improving closed-loop planning stability 15% and reducing off-road trajectories across simulated driving scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built **testable**, object-oriented Java and Python modules with Singleton and Factory patterns, covered by 20+ automated unit and integration suites run nightly in Jenkins across three operating systems.
- Optimized large-scale API integrations with pagination, caching, and field filtering, improving response times 30% and **scalability** for low-bandwidth users while cutting technical debt 20% in Agile teams.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **JavaScript** React and Java Spring Boot platform with RESTful microservices and real-time WebSocket updates, syncing 2M+ records across locations for resource, billing, and lead workflows.
- Took part in **code review** and rollout across 6+ workflows, gating releases on Jest and Selenium tests in Jenkins CI/CD with Docker packaging, reaching 95%+ successful build validation.

Skills

Languages: Java, Python, JavaScript, TypeScript, C++, SQL, HTML, CSS
Web & Backend: React, Next.js, Spring Boot, Flask, FastAPI, Node.js, REST APIs, Microservices
AI & Data: PyTorch, Transformers, LLM Systems, Computer Vision, PostgreSQL, MySQL, MongoDB
Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Design Patterns, Debugging
Tools: Git, Jenkins, CI/CD, Docker, AWS, Jest, Selenium, PyTest, Code Reviews, Agile, Linux, GitHub Actions

Projects

AI Template Generator for Social Posts | *React, TypeScript, Node.js, LLM APIs, Canvas*

- Designed a **React** editor that turns a short prompt into editable social-media layouts, with AI-suggested copy and color palettes, layer controls, and **export** to PNG and PDF.

PDF Q&A with Citations | *Python, FastAPI, Embeddings, FAISS, React*

- Designed a **PDF** assistant that splits uploaded documents into passages, answers questions with page citations using embedding search, and is checked with an **evaluation set** of known answers.

PiXMod Live | *Next.js, SAM, Stable Diffusion, CLIP, MongoDB, Stripe*

- Built an AI **image editing** SaaS with SAM segmentation and recoloring, Stable Diffusion inpainting and restoration, and CLIP search over 100K+ indexed image embeddings, raising premium conversion 18%.